

Mohammad Abbas Full Stack Developer

mhmd.y.abbass@gmail.com | 71255014 | Beirut, Lebanon | [Portfolio](#) | [LinkedIn](#) | [Github](#)

SUMMARY

Full Stack Developer with experience building web applications using React, Node.js, Go, and SQL. Focused on developing responsive user interfaces, designing REST APIs, and deploying production systems on GCP using Docker and CI/CD pipelines.

EXPERIENCE

Full Stack Developer, Big Data Specialist 12/2025 – Present | Remote

- Built a web platform using Next.js, Go, and MySQL to aggregate jobs, tech news, blogs, books, and GitHub repositories, delivering a unified content discovery experience with responsive interfaces for search, browsing, and dashboard-style navigation across large, multi-source datasets
- Integrated backend APIs into frontend workflows to enable dynamic, data-driven user experiences
- Implemented automated testing using Vitest and Playwright to ensure reliability across core user flows
- Deployed and maintained the application on GCP using Docker and CI/CD pipelines

Full Stack Developer, Vertex Partners 03/2025 – 11/2025 | Remote

- Built and maintained full-stack features for a high-traffic student housing platform using Next.js, Express.js, and PostgreSQL, delivering responsive interfaces for property listings, search, and management workflows
- Developed interactive map-based UI experiences enhanced with Framer Motion, while implementing REST APIs, database queries, and role-based access control for administrators, landlords, and students to ensure secure platform operations
- Improved application performance through rendering optimizations, code-splitting, and efficient data-fetching strategies, achieving Lighthouse scores above 90 and delivering features in agile development cycles

Full Stack Developer Intern, 3E Tech 12/2024 – 03/2025 | Beirut, Hybrid

- Developed full-stack features for LMS and property management platforms using React, TypeScript, NestJS, PostgreSQL, and Prisma, supporting core workflows across education and real estate systems
- Built modular frontend architecture with Redux Toolkit for complex state management, improving maintainability and scalability of large UI workflows
- Designed and optimized backend services and PostgreSQL queries using Prisma ORM, reducing API response time by ~25% through performance tuning and query optimization

Software Developer, Bracket Technologies 05/2024 – 08/2024 | Hazmieh, Lebanon

- Built a high-performance search and filtering system for flight and hotel listings, improving data retrieval efficiency and user query experience across large datasets
- Designed and implemented backend authentication and role-based access control, supporting secure multi-role workflows for users and administrators
- Contributed to scaling platform features in a collaborative engineering team, improving system reliability and maintaining consistent performance under increasing load

SKILLS

Programming Languages: JavaScript, TypeScript, Go, Python, SQL

Frontend: React, Next.js, Tailwind

Backend: Node.js (NestJS, Express), Gin, Flask

Databases: PostgreSQL, MySQL, Firebase

DevOps: Docker, GCP, GitHub Actions, CI/CD

Automated Testing: Vitest, Playwright

PROJECTS

MoDB

- Built a SQL-compatible relational database management system in Go with persistent disk storage, supporting multi-database and multi-table operations via a custom TCP-based query interface
- Designed and implemented core database internals including a SQL lexer, recursive descent parser, AST-based query planner, and execution engine for SELECT, INSERT, UPDATE, and DELETE operations
- Developed a custom storage engine using 4KB slotted-page architecture with binary persistence and schema storage, enabling durable data storage across restarts
- Implemented full relational constraints including PRIMARY KEY, FOREIGN KEY, UNIQUE, NOT NULL, and referential integrity with cascade protection
- Engineered a modular database pipeline (transport layer, SQL layer, execution layer, storage engine) closely modeled after production RDBMS architecture

Custom Programming Language

- Designed and implemented a custom programming language using symbol-based syntax instead of traditional keywords, exploring alternative language design paradigms
- Built a full parsing system using Nearley, including lexical analysis and syntax parsing for expressions, control flow, functions, and data structures
- Developed a transpiler that converts Mo code into executable JavaScript, enabling integration with the JavaScript runtime ecosystem
- Implemented core language features including variables, functions, loops, conditionals, and arrays
- Engineered a custom language grammar supporting symbolic constructs for assignments, function definitions, and control flow expressions

Job Match Agent

- Developed a data aggregation system scraping 10+ job platforms in parallel.
- Implemented semantic matching using embeddings + cosine similarity for intelligent job recommendations.
- Designed deduplication and historical tracking systems for data consistency.
- Built backend services in Python with scalable processing pipelines.

Rapid Express

- Engineered a globally-installed CLI tool to automate Express.js app scaffolding, enabling rapid generation of production-ready APIs with a single command.
- Designed a scalable folder architecture with built-in decorators and error-handling middleware, reducing initial developer setup time by 70%.
- Published on npm, achieving 1,200+ downloads and creating a valuable tool for the developer community.

EDUCATION

Bachelor in Management Information Systems,
Lebanese University

2021 – 2025 | Lebanon